Multi-Agent Forensic Reasoning for Generalizable Deepfake Video Detection

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Abstract

The malicious use of generative artificial intelligence to create highly realistic deepfake videos raises serious ethical concerns and poses substantial challenges to AI safety. However, existing deepfake video benchmarks provide limited coverage of recent synthesis methods and generally lack reliable fine-grained textual annotations. Meanwhile, conventional detectors and multimodal large language models (MLLMs), whether operating as a single model or relying on a single analytical perspective, often fail to capture subtle forgery artifacts, limiting their generalization to emerging AI-generated methods. To address these limitations, we introduce FaceVid-Forensics-100K, a large-scale deepfake video dataset comprising 100,000 videos and spanning 33 synthesis methods across face swapping, face reenactment, and entire-face synthesis, including recent generators such as Seedance 2.0. The dataset provides fine-grained textual annotations of visual observations and verdict-consistent forensic explanations, automatically synthesized through a multi-model aggregation and conflict-resolution pipeline powered by advanced MLLMs. Building on this benchmark, we propose a multi-agent forensic reasoning framework that employs four specialized domain-expert agents to independently analyze forgery cues from four perspectives: texture, lighting, motion, and physics. A judge agent then reconciles their reports to produce a final prediction together with an explanation. Extensive evaluations on out-of-domain test sets show that, despite being composed entirely of small open-source MLLMs, our framework outperforms all methods including closed-source GPT and Gemini models and ranks first across all reported metrics on this benchmark.

Introduction

The rapid advancement of generative artificial intelligence has made it increasingly easy to synthesize facial videos with high visual fidelity and temporal coherence. This progress creates unprecedented opportunities for creative industries, but also raises serious ethical and security concerns. Therefore, reliable deepfake video detection has become a critical task in digital media forensics. Yet the problem is no longer limited to identifying conspicuous blending boundaries or low-level generation artifacts. A new generation of video generators (Team Seedance et al. 2026; OpenAI 2025; Kuaishou 2024; Wan Team et al. 2025) can synthesize entire faces, maintain relatively stable identity and appearance across consecutive frames, and produce seemingly plausible

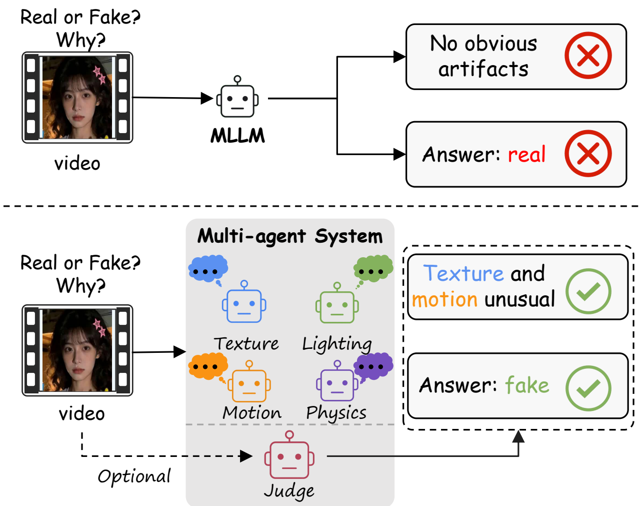


Figure 1: Framework comparison. A single MLLM often overlooks subtle forensic artifacts, which can lead to incorrect predictions. In contrast, our multi-agent framework employs specialized agents to examine the input video from four distinct forensic perspectives: texture, lighting, motion, and physics. A judge agent then aggregates their findings to produce a more reliable explanation and final prediction of whether the video under analysis is ultimately real or fake.

motion under diverse scenes. Consequently, detectors trained on earlier forgery techniques often suffer substantial performance degradation when confronted with unseen generation methods (Cheng et al. 2026a; Shen et al. 2025).

Conventional detectors (Wang et al. 2023b; Xu et al. 2024; Guo et al. 2025; Han et al. 2025; Peng et al. 2024; Yan et al. 2025) are typically small vision models that learn discriminative visual representations primarily from binary real-or-fake labels. Although these methods can achieve strong in-domain performance, they often rely on shortcut features specific to particular datasets or generators and therefore struggle to adapt to new forgery methods, identities, compression conditions, and data sources. Existing benchmarks (Rossler et al. 2019; Li et al. 2020c; Dolhansky et al. 2020; Jiang et al. 2020; Yan et al. 2024; Li et al. 2025) further amplify this problem: widely used datasets cover only a limited portion of the rapidly expanding synthesis landscape, and even some recent datasets that increase the number of videos typically provide only binary labels without specifying the visual evidence supporting each decision. Such coarse-grained supervision makes it difficult both to learn subtle, diverse, and previously unseen forgery cues and to determine whether a detector has acquired transferable forensic knowledge or merely fitted dataset biases. Moreover, authenticity decisions without supporting evidence limit the trustworthiness and auditability of detection systems in high-risk real-world settings.

Multimodal large language models (MLLMs) (Park et al. 2026; Li et al. 2026; Tan et al. 2026; Sun et al. 2025) offer a promising direction for addressing these limitations. With strong visual understanding and language-generation capabilities, MLLMs can analyze video content and describe suspicious facial details, inconsistencies across frames, and violations of common physical patterns. Nevertheless, directly asking a single MLLM whether a video is real or fake remains unreliable. A general-purpose model may overemphasize the most salient appearance cue, overlook weak evidence spread across different frames, or generate a plausible explanation that is inconsistent with its final verdict. More fundamentally, treating deepfake detection as a single holistic judgment ignores the distinct causes of different forgery traces: texture over-smoothing, inconsistent illumination, unstable motion, and violations of physical plausibility each require different forms of forensic knowledge and reasoning.

To address these limitations, we introduce the large-scale deepfake video benchmark FaceVid-Forensics-100K. The dataset contains 100,000 videos spanning 33 synthesis methods, including recent generators such as Seedance 2.0. It comprises 21,075 real and 78,925 fake videos across the major forgery categories of face swapping, face reenactment, and entire-face synthesis. Unlike existing datasets that provide only binary labels, FaceVid-Forensics-100K provides fine-grained textual supervision for every video along four dimensions: texture, lighting, motion, and physics. Specifically, multiple advanced open- and closed-source MLLMs first generate observations and judgments independently. An aggregation model then consolidates the evidence, resolves conflicts among the models, and produces a forensic explanation consistent with the final verdict. FaceVid-Forensics-100K therefore records not only authenticity labels but also the interpretable evidence supporting each decision, establishing a foundation for evidence-driven deepfake detection.

Building on this benchmark, we further propose a multi-agent forensic reasoning framework that decomposes deepfake detection into independent yet coordinated specialized analyses. Four domain-expert agents examine texture statistics, illumination consistency, temporal motion patterns, and physical plausibility, respectively. A judge agent then aggregates the expert reports, weighs mutually supporting or conflicting evidence, and outputs both an authenticity prediction and a concise forensic explanation. This decomposition encourages each agent to search systematically for a specific class of forgery traces, while the judge agent retains a global view of the video. When one type of forgery trace is weak or absent, the final decision can still rely on evidence corroborated across multiple perspectives, reducing the risk that a single salient cue dominates an erroneous prediction.

Extensive out-of-domain evaluations validate the effectiveness of the proposed approach. On the reported benchmark, the full system achieves 69.87% accuracy, 81.82% recall, and 53.28% F1, outperforming small vision models, general-purpose open- and closed-source MLLMs, and forensics-tuned MLLMs. Compared with the strongest single-model baseline, our approach improves F1 from 47.45% to 53.28%, an absolute gain of 5.83 percentage points. Allowing the judge agent to access the video directly further increases F1 from 51.01% to 53.28%, indicating that the original visual information can effectively supplement the expert reports. These results show that explicit multi-perspective collaborative reasoning provides a more reliable basis for generalizable deepfake video detection than holistic judgment by a single MLLM.

Related Work

Deepfake Video Detection

Early deepfake benchmarks primarily focused on a small number of face-swapping (Li et al. 2020a) and reenactment methods (Thies et al. 2016; Thies, Zollhöfer, and Nießner 2019). FaceForensics++ (Rossler et al. 2019) established a standardized benchmark across several manipulation pipelines, Celeb-DF (Li et al. 2020c) introduced higher-quality face swaps, and DFDC (Dolhansky et al. 2020) substantially increased the number of subjects and videos. DeeperForensics-1.0 (Jiang et al. 2020) further incorporated real-world perturbations to evaluate robustness. As synthesis techniques (Zou et al. 2026; Cao et al. 2026) diversified, DF40 (Yan et al. 2024) and Celeb-DF++ (Li et al. 2025) broadened coverage across multiple face-forgery paradigms. These face-centric datasets have driven progress in scale, realism, and manipulation diversity, but their supervision remains predominantly binary. They therefore provide limited guidance about which visual evidence supports an authenticity decision. In contrast, FaceVid-Forensics-100K contains 100,000 face-centric videos spanning 33 synthesis methods, including recent systems such as Seedance 2.0, and augments binary authenticity labels with dimension-specific forensic observations and verdict-consistent explanations.

Most conventional detectors learn discriminative representations from binary labels. Some methods target manipulation traces such as blending boundaries (Li et al. 2020b), gaze behavior (Peng et al. 2024), or lip motion (Haliassos et al. 2021); others model temporal coherence and spatiotemporal inconsistency (Zheng et al. 2021; Wang et al. 2023b; Xu et al. 2024; Guo et al. 2025). Recent approaches improve transfer by adapting foundation-model features or suppressing generator-specific directions (Han et al. 2025; Yan et al. 2025; Cheng et al. 2026b). Despite these advances, such small vision models rely on binary supervision and can overfit specific artifacts, limiting generalization.

MLLMs make it possible to formulate deepfake detection as evidence-grounded visual reasoning rather than opaque binary classification. EDVD-LLaMA (Sun et al. 2025) adapts an MLLM to explain manipulated facial videos, VidGuard-R1 (Park et al. 2026) jointly improves detection and ex-

Dataset	Forgery Coverage		Video Scale			Textual Labels	
	#Synth. Methods	Latest Fake	Real	Fake	Total	Obs.	Exp.
DeepfakeDetection	5	–	363	3,068	3,431	×	×
Celeb-DF v2	1	VAE (2014)	590	5,639	6,229	×	×
DeeperForensics-1.0	1	DF-VAE (2020)	**50,000**	10,000	60,000	×	×
DF40†	23	HeyGen (2024)	716	28,837	29,553	×	×
Celeb-DF++	22	FLOAT (2025)	590	53,196	53,786	×	×
FaceVid-Forensics-100K	**33**	**Seedance 2.0 (2026)**	21,075	**78,925**	**100,000**	✓	✓

Table 1: Comparison of representative deepfake video datasets. † Only the video-based subsets of DF40 are counted. Obs. and Exp. stand for Observation and Explanation, respectively.

planation through reinforcement learning, Skyra (Li et al. 2026) grounds reasoning in annotated visual artifacts, and VideoVeritas (Tan et al. 2026) combines question–answer supervision with preference and perception-oriented reinforcement learning. However, single-model inference may overlook weak cues or let one artifact bias the verdict. Our framework instead enables explicit multi-perspective collaborative reasoning: four specialized agents produce dimension-specific evidence, which a judge reconciles into the final explanation and verdict.

Multi-Agent Systems

Multi-agent systems coordinate specialized decision makers through communication and information exchange. Recent LLM-based systems use debate, critique, and iterative collaboration to improve factuality and reasoning (Du et al. 2024; Wu et al. 2024; Gao et al. 2025; He et al. 2025), while learning-based approaches optimize the interaction policies of collaborating agents (Wu et al. 2026; Feng et al. 2026; Zhao et al. 2026; Qiao et al. 2026). Multi-agent designs have also been extended to multimodal tasks: LongVideoAgent (Liu et al. 2026) coordinates grounding and visual agents for long-video understanding, and UniShield (Huang et al. 2026) routes specialized forensic tools for image manipulation detection and localization. Inspired by role-specialized multi-agent collaboration, we decompose deepfake video detection into four independently forensic perspectives, with each agent generating observation evidence. A judge then reconciles the evidence, establishing an explicit multi-perspective collaboration mechanism.

FaceVid-Forensics-100K

Dataset Overview

FaceVid-Forensics-100K is a large-scale deepfake video dataset comprising 100,000 face-centric videos. It covers 33 synthesis methods across major forgery categories, including face swapping, face reenactment, and entire-face synthesis. Beyond binary authenticity labels, the dataset provides fine-grained textual annotations of visual observations across four forensic dimensions—texture, lighting, motion, and physics—as well as verdict-consistent explanations. As shown in Figure 2, it serves as a foundation for training and evaluating evidence-driven deepfake detection systems.

Collection and Processing. Our dataset originates from two types of sources: existing general video forgery datasets and videos directly collected from the internet or synthesized via recent generative models. Specifically, we collect videos from AIGVDBench (Ma et al. 2026), GenVidBench (Ni et al. 2026), ViF-Bench (Li et al. 2026), and FaceForensics++ (FF++) (Rossler et al. 2019), yielding approximately 442,000, 6,780,000, 3,000, and 5,000 videos, respectively. In addition, we crawl 2,901 videos generated by Seedance 2.0 (Team Seedance et al. 2026) and other recent models from the internet, retaining 577 videos after manual screening. Following data collection, we apply a face detection pipeline to filter the videos frame-by-frame, retaining only frames containing faces and discarding videos without valid facial regions. After this preprocessing step, we obtain 1,158,585 valid face-centric videos, comprising 30,240 real videos and 1,128,345 fake videos, with a total duration of 436.56 hours.

To ensure a balanced distribution, we first deduplicate the real videos based on their YouTube IDs, which reduces the real samples from 30,240 to 21,075. For the fake videos, four synthesis models—`ModelScope` (Wang et al. 2023a), `Pika` (Pika 2024), `Text2Video-Zero` (Khachatryan et al. 2023), and `VideoCrafter2` (Chen et al. 2024)—dominate the collection. We evaluate these fake videos using the AltFreezing (Wang et al. 2023b) detector and prioritize retaining samples with lower scores (i.e., those harder to distinguish). Through this strategy, the number of videos from these four models is reduced from 235,007, 75,471, 435,210, and 376,471 to 11,958, 15,163, 10,432, and 32,690, respectively. This decreases their total count from 1,122,159 to 70,243. Combined with the other retained fake videos, the total number of fake videos is reduced from 1,128,345 to 78,925. The final curated dataset contains 100,000 videos, consisting of 21,075 real videos and 78,925 fake videos. These forgeries cover face swapping (FS), face reenactment (FR), and entire-face synthesis (EFS), which encompasses text-to-video (T2V), image-to-video (I2V), text-and-image-to-video (TI2V), and video-to-video (V2V) generation paradigms.

Training and Evaluation. We divide the dataset into training, in-domain test, and out-of-distribution (OOD) test sets. The training and in-domain test sets share the same families of synthesis methods. To rigorously evaluate cross-generator

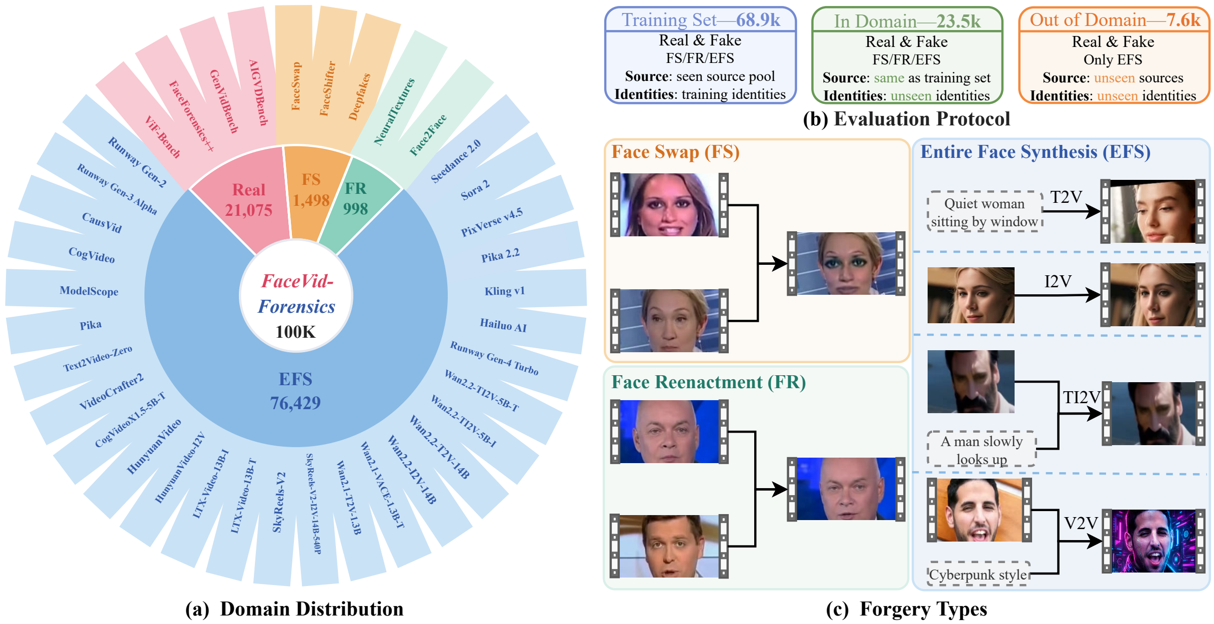


Figure 2: Overview of FaceVid-Forensics-100K.

generalization, we assign videos from 20 completely unseen (held-out) EFS generators exclusively to the OOD test set, along with a portion of the real videos. As detailed in Appendix A, these held-out generators include recent systems such as CogVideoX1.5, HunyuanVideo, LTX-Video, SkyReels, Wan, Hailuo, Kling, Pika 2.2, PixVerse, Sora 2, and Seedance 2.0. In total, the training, in-domain test, and OOD test splits contain approximately 68.9K, 23.5K, and 7.6K videos, respectively.

Label Generation

To construct fine-grained textual supervision without relying on manual annotation, we employ an ensemble of five diverse MLLMs spanning three distinct paradigm types as annotators, as shown in Figure 3: (1) powerful closed-source MLLMs (GPT-4o (OpenAI 2024) and Gemini 3.5-Flash (Google DeepMind 2026)); (2) general-purpose open-source MLLMs (Qwen2.5-VL (Bai et al. 2025)); and (3) forensics-tuned domain-specific MLLMs (Skyra (Li et al. 2026) and VideoVeritas (Tan et al. 2026)). To synthesize their outputs into unified, high-quality textual supervision, we select DeepSeek-V4 Pro (DeepSeek-AI et al. 2026) as the central aggregator based on two key design considerations: first, label aggregation operates strictly over textual observation/explanation reports and ground-truth metadata, making a pure text LLM with strong logical reasoning capabilities optimal without requiring multimodal visual inputs; second, decoupling the aggregator from the annotator pool ensures architectural independence, preventing the aggregator from inheriting potential inductive biases or error patterns present in the visual annotator MLLMs.

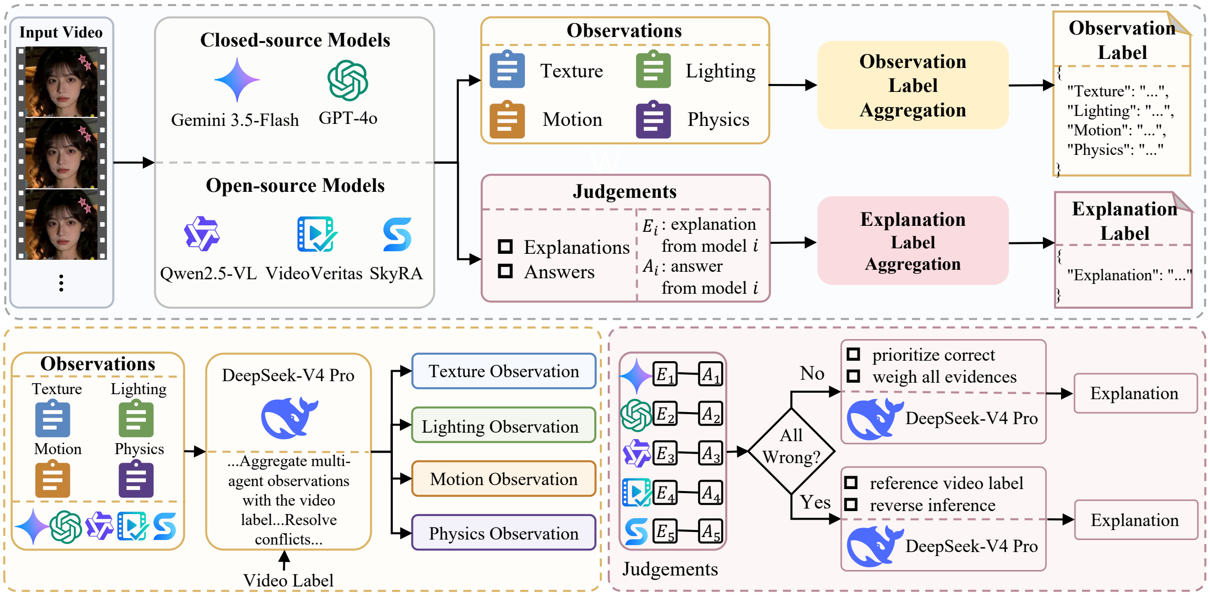
Observation Label. As illustrated in Figure 3(a), the five annotators independently analyze each video to generate detailed observations across the four forensic dimensions (Texture, Lighting, Motion, and Physics). The aggregator model (DeepSeek-V4 Pro) then integrates these multi-annotator outputs by merging the observations separately within each forensic dimension. Guided by the ground-truth video label to resolve cross-model contradictions and filter noise, the aggregator outputs a structured observation label containing consensus descriptions for each dimension without revealing the final authenticity verdict.

Explanation Label. In addition to dimensional observations, each annotator produces an authenticity verdict (real or fake) alongside an initial explanation rationale. As shown in Figure 3(b), the aggregator model consolidates these individual answer–explanation pairs into a single, verdict-consistent explanation label. This aggregation is strictly conditioned on the ground-truth video label: when correct predictions exist, the aggregator prioritizes and synthesizes explanations from correct annotators; if all annotators make incorrect predictions, it performs reverse inference by re-evaluating the merged observations against the ground-truth label. This ensures the final textual rationale is aligned with the correct authenticity verdict while eliminating individual model biases, noise, and hallucinations.

Multi-Agent Forensic Reasoning

Multi-Agent System

As shown in Figure 4, our framework decomposes deepfake video detection into independent forensic analyses coordinated by a hierarchical multi-agent system. For a given input video v, we uniformly sample a sequence of frames $X(v)$ and provide them to the system, which consists of four specialized observation agents and a central judge agent.



(a) Observation Label Aggregation (b) Explanation Label Aggregation

Figure 3: Pipeline of observation and explanation label generation. (a) DeepSeek-V4 Pro aggregates independent observations separately within each forensic dimension. (b) It combines answer–explanation pairs into a verdict-consistent explanation label.

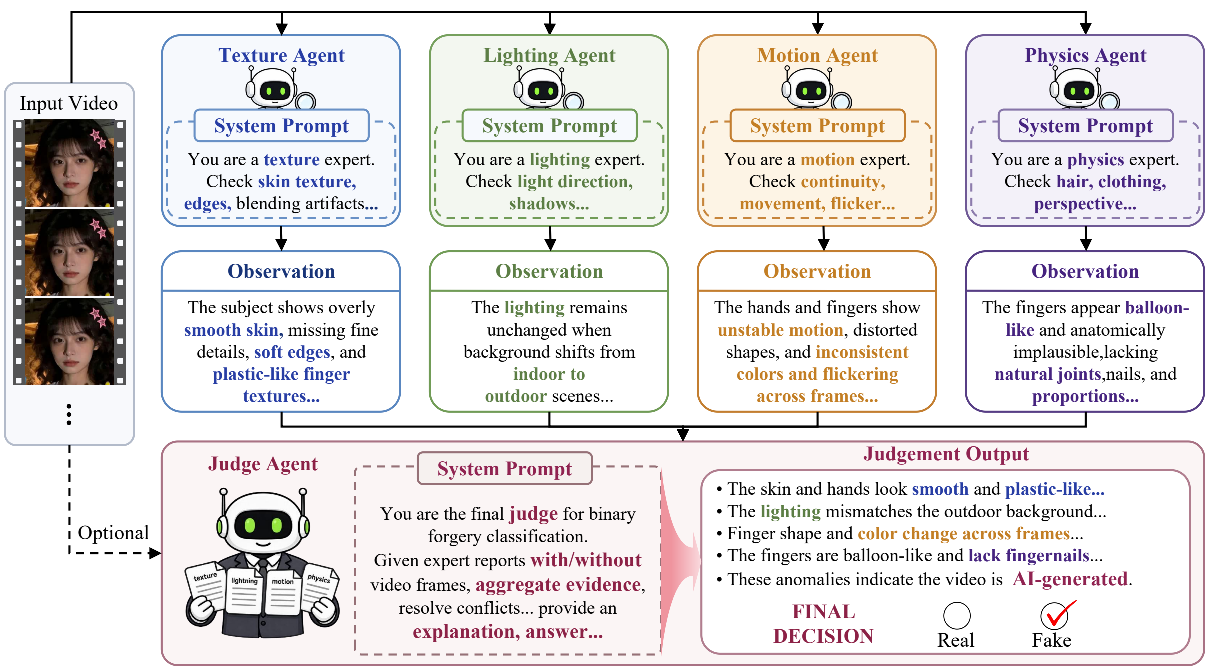


Figure 4: Overview of the proposed multi-agent forensic reasoning framework.

Observation Agents. The four observation agents focus on distinct aspects of forgery-related artifacts: texture, lighting, motion, and physics. Each observation agent $\mathcal{A}_d$ independently receives the same sampled frames $X(v)$ along with a dimension-specific prompt, and produces a textual observation $\widehat{O}^d$ without predicting the final authenticity. We instantiate all agents using MLLMs. Their system prompts instruct each agent to search systematically for its assigned cue type: the texture agent evaluates skin details and blending boundaries, the lighting agent assesses illumination and shadow consistencies, the motion agent tracks temporal instability, and the physics agent considers anatomical and geometric plausibility. The complete prompts for these agents are detailed in Appendix F.

Judge Agent. The judge agent acts as the central reasoning hub. It receives the textual observations generated by the four observation agents and, optionally, the sampled video frames $X(v)$. Its prompt instructs it to weigh mutually supporting or conflicting evidence from the experts and output a binary real-or-fake answer alongside a concise, verdict-consistent explanation. This explicit multi-perspective collaboration ensures that decisions rely on corroborated evidence rather than being overly influenced by a single salient artifact. The exact prompt template for the judge agent is provided in Appendix F.

Training Pipeline

We train the proposed multi-agent framework in two sequential stages: supervised fine-tuning (SFT) for all agents, followed by group relative policy optimization (GRPO) (Shao et al. 2024) to refine the judge agent's decisions.

Supervised Fine-Tuning. During SFT, the four observation agents are trained independently using the aggregated, dimension-specific observation labels $O^d_\star = (o^d_1, \ldots, o^d_{L_d})$ collected during the dataset construction process. For each dimension $d \in \mathcal{D}$, the corresponding agent $\mathcal{A}_d$ is optimized via autoregressive language modeling:

$$\mathcal{L}^d_{\text{obs}}(\theta_d) = -\frac{1}{N_d} \sum_{(v,O^d_\star)\in\mathcal{T}} \sum_{t=1}^{L_d} \log p_{\theta_d}\left(o^d_t \mid X(v), I_d, o^d_{<t}\right), \tag{1}$$

where $\mathcal{T}$ is the training set, θ_d denotes the agent's parameters, I_d is the instruction, and N_d is the number of response tokens. After training the observation agents, we apply them to the training set to generate their intermediate outputs $R(v) = \text{Concat}_{d\in\mathcal{D}}(\widehat{O}^d)$. The judge agent is then trained on these concatenated observations and the visual input $V_c(v)$ (where $c \in \{0, 1\}$ indicates the presence of video frames), taking $Q_c(v) = (R(v), V_c(v))$ as input. The judge agent learns to produce the target explanation and verdict $Z_\star = (z_1, \ldots, z_{L_J})$ by minimizing:

$$\mathcal{L}^c_{\text{J}}(\phi_c) = -\frac{1}{N_{\text{J}}} \sum_{(v,Z_\star)\in\mathcal{T}} \sum_{t=1}^{L_J} \log p_{\phi_c}(z_t \mid Q_c(v), z_{<t}). \tag{2}$$

Policy Optimization for Decision Refinement. Following SFT, we apply GRPO to further align the judge agent's final decision policy, keeping the four observation agents frozen. For each input $Q_c(v)$, the judge agent samples a group of G candidate responses $\{\widetilde{Z}_g\}_{g=1}^G$. A binary accuracy reward $r_g \in \{0, 1\}$ is assigned based on whether the candidate correctly predicts the ground-truth label. We compute the relative advantage A_g by normalizing the rewards within the group:

$$A_g = \frac{r_g - \overline{r}}{\sigma_r + \epsilon}. \tag{3}$$

GRPO then updates the judge agent's parameters to maximize this advantage using a clipped surrogate objective, complemented by a KL-divergence penalty against the frozen SFT policy to maintain explanation quality. This stage refines the final classification accuracy without altering the specialized evidence extracted by the observation agents.

Experiments

Experimental Setup

Implementation Details. To ensure a fair comparison with Skyra (Li et al. 2026), we adopt Qwen2.5-VL-7B (Bai et al. 2025) as the base model for all agents. All training is implemented using ms-swift (Zhao et al. 2025). We train each component for one epoch using LoRA (Hu et al. 2022) and the AdamW optimizer (Loshchilov and Hutter 2018), with learning rates of 1×10^{-4} for supervised fine-tuning and 5×10^{-6} for GRPO. For GRPO, we sample eight responses per prompt. Code is available in the supplementary material.

Evaluation Protocols. We conduct the evaluation on the OOD split set, containing 5,716 real videos and 1,920 fake videos from 20 generators excluded from training. For small vision models, we retrain each method on FaceVid-Forensics 100K using its official code and data preprocessing pipeline. Off-the-shelf general MLLMs and forensics-tuned MLLMs are evaluated using their released checkpoints or official APIs. For models that support an explicit thinking mode, we disable it during inference. Following Skyra (Li et al. 2026), we use accuracy (Acc), F1 score, and recall as the evaluation metrics.

Comparison with State-of-the-Art Methods

As shown in Table 2, we compare our framework with small vision models, general-purpose open- and closed-source MLLMs (Qwen Team 2026; Xiaomi MiMo Team 2026; OpenAI 2024; Google DeepMind 2026), and forensics-tuned MLLMs (Li et al. 2026; Tan et al. 2026) on the OOD test set. Our framework achieves the strongest overall performance when the judge agent receives both the observation reports and sampled video frames. The text-only judge agent also ranks second, showing that the observation agents provide effective forensic evidence, while direct access to the video further improves OOD generalization. In-domain results, per-generator OOD results, qualitative comparisons, and evaluation of textual explanation quality are available at Appendix B.

Ablation Studies

Contribution of Each Observation Agent. As shown in Table 3, combining all four observation agents gives the best F1 score under both judge agent configurations. The advantage is clearer without direct video input, indicating that the four perspectives provide distinct evidence. When frames are also available, the gap narrows because the judge agent can recover part of the missing visual information directly.

Effect of Training with SFT and GRPO. To isolate the contribution of each trained component, Only-Observation trains the observation agents while keeping the Judge untrained, whereas Only-Judge trains the Judge using outputs from untrained observation agents. As shown in Table 4, training either component improves the training-free system,

Method	Acc	Recall	F1
Small Vision Models			
DFGaze *(TIFS'24)*	56.67	17.92	27.24
DFD-FCG *(CVPR'25)*	61.25	29.22	39.16
Effort *(ICML'25)*	63.92	34.79	44.76
TALL++ *(IJCV'24)*	63.47	42.97	45.07
TFCU *(CVPR'25)*	64.28	33.44	45.20
Open-source MLLMs			
Qwen2.5-VL-7B	35.94	16.51	13.24
InternVL3.5-8B	37.87	17.60	14.53
MiMo-V2.5 (310B-A15B)	49.85	15.00	18.69
Qwen3.6-35B-A3B	53.17	50.05	35.72
Closed-source MLLMs			
GPT-4o (2024)	57.63	40.36	37.55
GPT-5-mini (2025)	59.31	38.70	39.00
Gemini-2.5-Pro (2025)	63.78	75.29	47.45
Gemini-3.5-Flash (2026)	63.34	58.75	46.22
Forensics-tuned MLLMs			
Skyra *(CVPR'26)*	60.50	30.05	38.30
VideoVeritas *(ICML'26)*	57.87	<u>78.96</u>	43.22
Multi-Agent System			
Ours (w/o Video)	<u>67.41</u>	65.00	<u>51.01</u>
Ours (w/ Video)	**69.87**	**81.82**	**53.28**

Table 2: Comparison with state-of-the-art deepfake video detectors on the OOD test set. The best and second-best available results are highlighted in bold and underlined.

Observation	Ours (w/o Video)			Ours (w/ Video)		
	Acc	Recall	F1	Acc	Recall	F1
Texture	58.39	**97.40**	44.53	<u>66.90</u>	**87.86**	<u>50.39</u>
Lighting	<u>59.60</u>	<u>94.22</u>	<u>45.13</u>	65.87	85.16	49.49
Motion	54.70	75.31	40.54	63.15	83.39	47.20
Physics	56.43	91.82	43.02	62.85	87.45	47.11
All four	**63.61**	82.24	**47.53**	**67.03**	<u>87.76</u>	**50.49**

Table 3: Comparison of individual observation agent outputs and their combination on the OOD test set. The best and second-best results within each judge agent configuration are highlighted in bold and underlined, respectively.

jointly training the observation agents and judge is more effective than training either alone, and GRPO provides a further gain. These results confirm that specialized observation learning, evidence reconciliation, and policy optimization each contribute to the final generalizable performance. Relevant training dynamics are provided in Appendix G. Furthermore, we investigate the impact of different combinations of MLLMs for the observation and judge agents in Appendix D and the impact of model parameter scale in Appendix E.

Reasoning Strategies across MLLMs. We compare our multi-agent design with direct prediction, chain-of-thought (CoT) prompting, and three multi-turn variants using frozen MLLMs, as shown in Table 5. Specifically, all multi-turn baselines decompose the forensic reasoning into 5 sequential dialogue turns (4 observation turns for texture, lighting, motion, and physics, followed by a final decision turn), but differ in frame delivery across turns: *Multi-turn* inputs video frames only in the first turn, *Multi-turn-Obs* provides frames across all four observation turns (turns 1–4), and *Multi-turn-All* feeds frames continuously across all 5 turns. Across both backbones, our framework performs best overall. This result shows that the improvement comes from independent evidence collection and Judge-based reconciliation, rather than from longer prompts, additional dialogue turns, or repeated access to the video frames. Detailed implementations, efficiency analyses, and qualitative comparisons of these reasoning strategies are provided in Appendix C.

Stage	w/o Video			w/ Video		
	Acc	Recall	F1	Acc	Recall	F1
Training-Free	45.06	65.05	33.53	42.29	54.06	30.07
+SFT (Only-Obs)	50.63	99.58	40.46	50.64	99.84	40.48
+SFT (Only-Judge)	51.42	99.95	40.87	64.29	95.78	48.39
+SFT (Joint)	63.61	<u>82.24</u>	47.53	67.03	**87.76**	50.49
+SFT+GRPO	<u>67.41</u>	65.00	<u>51.01</u>	**69.87**	81.82	**53.28**

Table 4: Effect of training with SFT and GRPO on the OOD test set. The best and second-best results are highlighted in bold and underlined, respectively.

Strategy	Qwen2.5-VL-7B			InternVL3.5-8B		
	Acc	Recall	F1	Acc	Recall	F1
Single	35.94	16.51	13.24	37.87	17.60	14.53
CoT	30.93	15.31	11.29	41.90	2.81	3.53
Multi-turn	33.98	15.26	11.92	40.53	41.56	25.85
Multi-turn-Obs	33.46	11.61	9.50	38.12	40.26	24.34
Multi-turn-All	33.88	13.87	11.05	37.22	38.80	23.48
Ours (w/o Video)	**45.06**	**65.05**	**33.53**	**45.40**	<u>46.46</u>	<u>29.81</u>
Ours (w/ Video)	<u>42.29</u>	<u>54.06</u>	<u>30.07</u>	<u>45.29</u>	**49.22**	**30.52**

Table 5: Training-free comparison of reasoning strategies across MLLMs on the OOD test set. Best and second-best results within each backbone are highlighted in bold and underlined, respectively.

Conclusion

We presented FaceVid-Forensics-100K, a large-scale deepfake video benchmark with broad coverage of recent synthesis methods and fine-grained forensic annotations. We further proposed a multi-agent forensic reasoning framework that performs collaborative analysis from four complementary forensic perspectives and produces both authenticity predictions and explanations. Extensive experiments demonstrate that our approach consistently outperforms existing vision-based detectors and MLLMs on out-of-domain benchmarks, highlighting the effectiveness of multi-perspective collaborative reasoning for generalizable deepfake video detection.

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